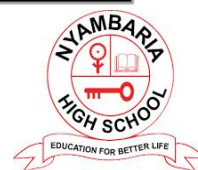
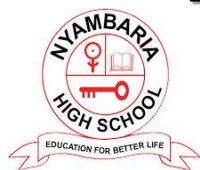


NYAMBARIA BOYS' 2025 PREDICTION



KCSE PROJECTION EXAM

COMPUTER STUDIES

451/2

PAPER 2

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education

INSTRUCTIONS TO CANDIDATES

- a)** *Indicate your name and index number at the right hand corner of each printout*
- b)** *Write your name and index number on the CD/removable storage medium provided*
- c)** *Write the name and version of the software used for each question attempted in the answer sheet provided*
- d)** *Answer all the questions, All questions carry equal marks*
- e)** *Passwords should not be used while saving in the CD/removable storage Medium*
- f)** *Marked printout of the answers on the sheet*
- g)** *Hand in all the printouts and the CD/removable storage medium used*

1. (a)(i) The extract below shows a spreadsheet used to compute the toll charges for a highway based on the type of vehicle, tonnage and charge per kilometer for usage.

<u>HIGHWAY TOLL CHARGES</u>						
Registration	Vehicle Type	Weight	Distance	NormalCharge	Penalty Charge	Total
KCY 789 M	PickUp	6	12			
KCR 769 L	Car	4	40			
KCF 724 C	PickUp	6	32			
KCM 737 N	Truck	12	25			
KCA 745 W	Lorry	20	28			
KCP 756 H	Truck	10	12			
KCU 778 J	Car	4	8			
KCZ 701 A	PickUp	8	25			
KCB 781 E	Car	6	4			
KCV 743 H	PickUp	4	20			
KCQ 735 X	Truck	8	32			
KCT 721 K	Lorry	10	25			
KCD 792 V	Truck	12	28			
KCZ 784 P	Car	6	12			
KCB 756 C	Truck	10	8			
KCE 734 D	Car	4	25			
KCF779 E	PickUp	6	32			
KCG 700 F	Lorry	12	25			
KCH 723 K	Truck	20	28			
KCJ 711 W	PickUp	10	12			
KCR 712 D	Car	4	8			
KCD 774 B	PickUp	8	25			
KCS 756 M	Truck	6	4			

KCA 745 W	Car	4	20			
			<u>Total</u>			

- (ii) Create a workbook and save the workbook as toll. **(2 Marks)**
 (iii) Fill the data in the worksheet1 and rename the worksheet as tollOriginal. **(14 Marks)**

(b)(i) The NormalCharge column is calculated based on the table below.

Vehicle Type	Max AllowedWeight(Tns)
PickUp	6
Car	4
Truck	8
Lorry	10

- (ii) The PenaltyCharge column is calculated based on the table below. The penalty is based on any weight above Maximum allowed weight for a vehicle type for every kilometer of the the usage.
(8 Marks)

Vehicle Type	Penalty Charge (Ksh)per Km
PickUp	10
Car	5
Truck	15
Lorry	20

- (iii) The TotalCharge is based on summation of NormalCharge and PenultyCharge . Create a column TotalCharge and use a function to Calculate the Total Charge **(2 Marks)**
 (iv) Create the Running Totals for Normal Charge, Penulty Charge And Totalcharge **(4 Marks)**
 (b)(i) Copy The data in the OriginalToll to another worksheet rename the workshhet as Sorted **(1 Mark)**
 (ii) Sort the Data is ascending order of Vehicle type Sorted worksheet. **(4 Marks)**
 (iii) Create subtotals based on the vehicle Type **(4 Marks)**
 (iv) Draw a column chart based on The Vehicle Type subtotals and Total Charge **(8 Marks)**

2. a) Create a database called **Aberdare bottles ltd** and create the following tables **(15 marks)**

Table 1: Employee

Employee_ID	EmployeeName	Department	YearOfEmployment
101	Kibet Arap Kamau	Human resource	1985

102	Janet Atieno	Procurement	1990
450	Kimani Koigu	Accounts	2000
891	Moraa Kerubo	Human resource	2010

Table 2: Sales

ProductName	Employee_ID	ProductID	SalesAmount	Salary
Tea leaves	101	Xc101	5000	
cocoa	102	Xp105	15500	
coffee	450	Xvb11	9500	
Chocolate	891	X56po	30000	

Table 3: Department

Employee_ID	Department_Name	HeadOfDepartment	NoOfEmployees
101	Human resource	B.N. Komu	52
102	Procurement	J.K. Wanjiru	12
450	Accounts	P.G. Otindo	20
891	Human resource	M.M. Jerotich	10

- i) Create **relationship** among the tables (2 marks)
- ii) Create **three** input screens (**forms**) and use them to enter the data into the tables above (6 marks)
- iii) Create a query called **Start_K** and use it to display **EmployeeName** that start with letter **K** (3 marks)
- iv) Display the **no of years** an employee has worked given that the current year is 2018. Save **the report** as **AGE**. (3 marks)
- v) Create a query called **Yote** to display the following fields (2 marks)
 - **Employee_ID**
 - **EmployeeName**
 - **Department_Name**
 - **ProductName**
 - **Salary**
 - **HeadOfDepartment**
- vi) Copy **Yote** query (in v above) and save the new query as **MPYA:-** (1 marks)
 Use **MPYA** query to
 - ✓ Calculate the salary given that: salary is 10% of the **SalesAmount** (2 marks)
 - ✓ Display salary in ascending order (2 marks)
 - ✓ Display employees from **human resource** department whose **SalesAmount** is **greater than 12000**. (2 marks)
- vii) Create a form called **AberdareForm** using **Yote** query (in v above) and use it to answer the questions below:-
 - ✓ Count no of employees (2 marks).
 - ✓ Add a title of the form as “**Aberdare bottles ltd-2018**” (2 marks)

✓ Insert **date and time** on the form header use (=NOW())

(2 marks)

viii) **Print** Age, Sales table, and AberdareForm

(3 marks)